Name: Praju Maharjan

Group: L5CG9

Tutorial - 5

1. Create a user defined datatype.

Ans :

Source Code :

```
#include<stdio.h>

struct Student{
 int roll;
 char grade;
}std;

int main(){
 printf("Size of structure : %ld.",sizeof(std));

 printf("\n");
 return 0;
}
```

Output :

```
Size of structure : 8.
```

2. Create a structure variable.

Ans:

Source Code :

```c
#include<stdio.h>

struct Student{
 char name[50];
 int level;
 char group[6];
};

int main(){
 struct Student s1={"Praju Maharjan",5,"L5CG9"};

 printf("Name : %s\n",s1.name);
 printf("Level : %d\n",s1.level);
 printf("Group : %s\n",s1.group);

 printf("\n");
 return 0;
}
```

Output :

```
Name : Praju Maharjan
Level : 5
Group : L5CG9
```

3. Use of typedef keyword in c.

Ans :

Source Code :

```c
#include <stdio.h>

typedef struct {
 char name[50];
 int level;
 char group[6];
}Student;

int main(){
 Student s1={"Praju Maharjan",5,"L5CG9"};

 printf("Name : %s\n",s1.name);
 printf("Level : %d\n",s1.level);
 printf("Group : %s\n",s1.group);

 printf("\n");
 return 0;
}
```

Output :

```
Name : Praju Maharjan
Level : 5
Group : L5CG9
```

4. Create a structure for student(roll,name,address,mark) and access the members.

Ans :

Source Code :

```
#include <stdio.h>
#include <string.h>

struct Student{
 int roll;
 char name[50];
 char address[50];
 int marks;
};

int main(){
 struct Student s1;

 s1.roll=16;
 strcpy(s1.name,"Praju Maharjan");
 strcpy(s1.address,"Kathmandu");
 s1.marks=87;

 printf("Roll No. : %d \n",s1.roll);
 printf("Name : %s \n",s1.name);
 printf("Address. : %s \n",s1.address);
 printf("Marks. : %d \n",s1.marks);

 printf("\n");
 return 0;
}
```

Output :

```
Roll No. : 16
Name : Praju Maharjan
Address. : Kathmandu
Marks. : 87
```

5. Create a structure for student(roll,name,address(province,district,city,ward),mark) and access the members.

Ans :

Source Code :

```c
#include <stdio.h>

struct Address{
 char province[20];
 char district[20];
 char city[20];
 int ward;
};

struct Student{
 int roll;
 char name[50];
 struct Address address;
 int marks;
};

int main(){
 struct Student s1={16,"Praju Maharjan",{"Bagmati","Kathmandu","Kathmandu",14},87};

 printf("Roll No. : %d \n",s1.roll);
 printf("Name : %s \n",s1.name);
 printf("Full Address : \n\tProvince : %s\n\tDistrict : %s\n\tCity : %s\nWard : %d\n",s1.address.province,s1.address.district,s1.address.city,s1.address.ward);
 printf("Marks. : %d \n",s1.marks);

 printf("\n");
 return 0;
}
```

Output :

```
Roll No. : 16
Name : Praju Maharjan
Full Address :
        Province : Bagmati
        District : Kathmandu
        City : Kathmandu
Ward : 14
Marks. : 87
```

6. Array of structure.

Ans :

Source Code :

```c
#include <stdio.h>
#include <string.h>

struct Student{
 int roll;
 char name[50];
 char address[50];
 int marks;
};

int main(){
 struct Student s[2];

 s[0].roll=16;
 strcpy(s[0].name,"Praju Maharjan");
 strcpy(s[0].address,"Kathmandu");
 s[0].marks=87;

 s[1].roll=17;
 strcpy(s[1].name,"Prajwal Shrestha");
 strcpy(s[1].address,"Lalitpur");
 s[1].marks=78;

 for(int i=0;i<=1;i++){
  printf("Roll No. : %d \n",s[i].roll);
  printf("Name : %s \n",s[i].name);
  printf("Address. : %s \n",s[i].address);
  printf("Marks. : %d \n",s[i].marks);
 }

 printf("\n");
 return 0;
}
```

Output :

```
Roll No. : 16
Name : Praju Maharjan
Address. : Kathmandu
Marks. : 87
Roll No. : 17
Name : Prajwal Shrestha
Address. : Lalitpur
Marks. : 78
```

7. Create a structure named student that has a name, roll number and marks as members. Assume appropriate types and size of members. Use this structure to

read and display records of 10 students. Create two functions: One is to read information of students and other to display the information.

Ans :

Source Code :

```
#include<stdio.h>
#include<string.h>

typedef struct{
 int roll_number;
 char name[20];
 int marks;
}Student;

 Student read_info(){
 int roll_no,marks;
 char name[20];
 Student s;

 printf("Enter Roll No.,Name and marks seperated by commas : ");
 scanf("%d,%[^,],%d",&roll_no,name,&marks);

 s.roll_number=roll_no;
 strcpy(s.name,name);
 s.marks=marks;

 return s;
}

void display_info(Student s){
 printf("\nRoll No. : %d\n",s.roll_number);
  printf("Name : %s\n",s.name);
  printf("Marks : %d\n",s.marks);
}

int main(){
 Student s[10];

 for(int i=0;i<10;i++){
  s[i]=read_info();
 }

 for(int i=0;i<10;i++){
  display_info(s[i]);
 }

 printf("\n");
 return 0;
}
```

Output :

```
Enter Roll No.,Name and marks seperated by commas : 1,John,66
Enter Roll No.,Name and marks seperated by commas : 2,Arthur,55
Enter Roll No.,Name and marks seperated by commas : 3,Shawn,12
Enter Roll No.,Name and marks seperated by commas : 4,Charles,88
Enter Roll No.,Name and marks seperated by commas : 5,Susan,44
Enter Roll No.,Name and marks seperated by commas : 6,Karen,99
Enter Roll No.,Name and marks seperated by commas : 7,Molly,75
Enter Roll No.,Name and marks seperated by commas : 8,Bill,0
Enter Roll No.,Name and marks seperated by commas : 9,Javier,42
Enter Roll No.,Name and marks seperated by commas : 10,Jack,74

Roll No. : 1
Name : John
Marks : 66

Roll No. : 2
Name : Arthur
Marks : 55

Roll No. : 3
Name : Shawn
Marks : 12

Roll No. : 4
Name : Charles
Marks : 88

Roll No. : 5
Name : Susan
Marks : 44

Roll No. : 6
Name : Karen
Marks : 99

Roll No. : 7
Name : Molly
Marks : 75

Roll No. : 8
Name : Bill
Marks : 0

Roll No. : 9
Name : Javier
Marks : 42

Roll No. : 10
Name : Jack
Marks : 74
```

8. Create a user defined data type for storing 2D coordinate point. Take two points from user and calculate midpoint using function.

Ans :

Source Code :

```c
#include <stdio.h>

typedef struct{
 float x;
 float y;
}point;

point calculate_mid_point(point p1,point p2){
 point mid_point;

 mid_point.x=(p1.x+p2.x)/2;
 mid_point.y=(p1.y+p2.y)/2;

 return mid_point;
}


int main(){
 point p1,p2,mid_point;
 float x,y;

 printf("Enter the co-ordinates of the first point(x1,y1): ");
 scanf("%f,%f",&x,&y);
 p1.x=x;
 p1.y=y;

 printf("Enter the co-ordinates of the second point(x2,y2): ");
 scanf("%f,%f",&x,&y);
 p2.x=x;
 p2.y=y;

 mid_point=calculate_mid_point(p1,p2);
 printf("Mid point of co-ordinates (%f,%f) and (%f,%f) is (%f,%f).",p1.x,p1.y,p2.x,p2.y,mid_point.x,mid_point.y);

 printf("\n");
 return 0;
}
```

Output :

```
Enter the co-ordinates of the first point(x1,y1): 1,2
Enter the co-ordinates of the second point(x2,y2): 2,4
Mid point of co-ordinates (1.000000,2.000000) and (2.000000,4.000000) is (1.500000,3.000000).
```